



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can use ratio reasoning to convert between units within the same measurement system.

Language Objective: I can explain a conversion using the words unit ratio, convert, measurement system, and per.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Conversion factor

Unit ratio

Convert

Measurement system

Double number line

Ratio table



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The ratio that trades one unit for another is a

2

A ratio comparing an amount to exactly unit is a unit ratio.

3

To change a measurement to another unit without changing the amount is to

4

Inches and feet belong to the customary

 ; centimeters and meters to the metric one.

5

Two lines whose matching ticks show the same amount in two units form a

 .

6

A table whose every column holds the same comparison is a

 .

Write About the Math

How many cup-scoops of broth does the recipe need, and how do you know?

¿Cuántas tazas de caldo necesita la receta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

☐

Conversion factor

Factor de conversión

☐

Unit ratio

Razón unitaria

☐

Convert

Convertir

☐

Measurement system

Sistema de medidas

☐

Double number line

Recta numérica doble

Model: Find the scale factor (120 divided by $8 = 15$), then multiply each ingredient by it. — Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75 , 3 cups mozzarella becomes 45).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The unit ratio is ____ cups to 1 quart. Cups are ____ than quarts, so I ____ to convert. $3 \times 4 =$ ____ cups. The amount of broth did not change — only the ____ did.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- Mi afirmación es ____.*
- My evidence is ____, so ____.
- Mi evidencia es ____, entonces ____.*

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Conversion factor
2. **Notes 2:** Unit ratio
3. **Notes 3:** Convert
4. **Notes 4:** Measurement system
5. **Notes 5:** Double number line
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size — multiplying every time. Converting 36 inches to feet by multiplying gives $36 \times 12 = 432$ feet, taller than a 40-story building. Ask first which unit is smaller: it takes MORE small units to make the same amount, so going to a smaller unit multiplies, and going to a larger unit divides.*
8. **Try It 1:** — *Every unit ratio says how many of the smaller unit fit inside exactly 1 of the larger one.*
9. **Try It 2:** — *Conversions in this lesson stay inside one system — customary with customary, metric with metric.*